TECHNICAL & HSEQ RIG PORTFOLIO

DRILLMEC HH SERIES | EQUIPMENT INTEGRITY | DRILLING OPERATIONS

Pietro Drogo • pietro.drogo@gmail.com • +39 338 128 1629 • Italy & Brazil

PROFESSIONAL SCOPE

Technical portfolio focused exclusively on hands-on knowledge of Drillmec HH-series land rigs, drilling equipment, maintenance, fault diagnosis, well control and safe field execution.

Technical scope: Drillmec HH rigs • Drilling equipment • Troubleshooting • Maintenance • Well control • HSE

TECHNICAL SCOPE & FIELD PRACTICE

- Direct technical knowledge of hydraulic hoisting rigs throughout drilling, tripping, casing, cementing and well intervention operations, including equipment interfaces and operating limitations.
- Detailed understanding of rig mechanical assemblies, hydraulic circuits, electrical systems, PLC interfaces, inspection routines, preventive maintenance and corrective repairs carried out in the field.
- Practical verification of rig readiness, drilling equipment condition, downhole component traceability and compliance with operating procedures before critical operations.
- Continuous cooperation with HSE personnel to apply task-specific risk assessments, permits to work, toolbox talks, equipment isolation and safe maintenance procedures.
- Systematic analysis of equipment faults, recurrent failure modes and wear mechanisms, including differentiation between operational misuse and normal service fatigue.
- Technical oversight of rig systems (hydraulic hoist, pipe handling, top drive, HPUs, mud systems, BOP stack, choke manifold), understanding capacities, limits, and failure modes to guide maintenance teams and resolve issues.
- Delivery of safe, practical solutions for client-specific operational requests, including non-standard scenarios.

Geographies: Italy, Brazil, Peru, Chile, Argentina, Venezuela, India.

FIELD TECHNICAL PRACTICE

- Hands-on troubleshooting and repair support for top drives, mud pumps, BOP assemblies, accumulator systems and auxiliary equipment, with attention to safe return-to-service checks.
- Routine monitoring of equipment condition, operating hours, critical consumables and inspection records to support timely maintenance and prevent avoidable equipment failures.
- Regular crew instruction and practical drills on equipment use, well-control response and safe execution of critical rig activities.

ONSHORE RIG EXPERIENCE

Note: All my experience has been onshore. No offshore assignments.

Drillmec HH100 Series (Rigs HH102 – HH104 – HH106 – HH107 – HH108 – HH109)

Rig type: Hydraulic Hoisting Rigs (onshore)

TECHNICAL & HSEQ RIG PORTFOLIO

Power: 1,000 HP • Max hook load 100 ton-m

Depth capacity: up to 1,500 m (3 phases)

Typical hole sections: 17" – 12 ¼" – 8 ½"

Casing programs: 13 3/8" – 9 5/8" – 7" or 5 ½"

Main equipment & tools:

- Hydraulic top drive 400 HP
- 5" G105 Range III drill pipe
- Automatic slips, casing devices, spider tools
- Hydraulic power tong for makeup/breakout torque
- Drillmec 9T1000 triplex mud pumps (National type)

Projects: Multiple onshore wells, 3-phase drilling, development wells in Italy and South America.

Drillmec HH200 Series (Rigs HH201 – HH202 – HH203 – HH204)

Rig type: Hydraulic Hoisting Rigs (onshore, cyber generation)

Power: 1,400 HP • Max hook load 200 ton-m

Power engine: 4 CAT 3412 – 1200 hp

HPU: 2 electrical motor 550 HP each

AUTOMATIC PIPE HANDLER TO MANAGE DRILL PIPE

Depth capacity: 3,500 – 3,800 m (depending on contract)

Typical hole sections: 27" – 17" – 12 ¼" – 8 ½"

Casing programs: 24" – 13 3/8" – 9 5/8" – 7" or 5 ½"

Main equipment & tools:

- Cyber rig package / automated systems
- Hydraulic power tong, casing devices, spider tools
- Top drive (hydraulic)
- 3 Mud pumps 9T100 DRILLMEC and solids control adapted to depth programs

Projects: International assignments with different clients across several countries.

Drillmec HH300 Series (Rigs HH300 Venezuela – HH300-1 – HH300-2 India – HH302 Brazil)

Rig type: Hydraulic Hoisting Rigs (onshore, high capacity)

Hydraulic power pack: 1,150 kW (approx. 1,542 HP) • Max hook load: 272 metric tonnes

Prime movers: 4 × CAT 3512 – reported 1,470 HP

HPU: 2 × 575 kW electric motors

Mud pumps: 3 × Drillmec 12T1600 triplex – 1,600 HP each

Rated drilling depth: 13,000 ft (approx. 3,962 m), based on 3½ in drill pipe

Typical hole sections: 17" – 12 ¼" – 8 ½"

Casing programs: 13 3/8" – 9 5/8" – 7" or 5 ½"

TECHNICAL & HSEQ RIG PORTFOLIO

Main equipment & tools:

- Hydraulic top drive 600 HP
- 5" S135 Range III drill pipe
- Pipe handling system, automatic slips
- Hydraulic power tong for torque operations

Projects: Venezuela, India, Brazil – development wells, deep drilling campaigns.

ADVANCED RIG EQUIPMENT KNOWLEDGE, MAINTENANCE & DRILLING OPERATIONS

Hands-on technical expertise developed through international onshore drilling operations on Drillmec HH-series rigs. My approach combines detailed equipment knowledge, preventive maintenance, fault diagnosis, operational decision-making and crew training. I work closely with maintenance specialists while retaining direct technical oversight of equipment condition, failure causes, repair feasibility and safe return to service.

Drillmec Hydraulic Top Drive Systems

Detailed familiarity with mechanical assemblies, hydraulic actuation and control circuits, gear-dependent torque/power characteristics and operating limitations. Able to diagnose recurrent faults, distinguish indications of misuse from fatigue-related damage and assess the severity of failures and their operational consequences. Direct experience planning and supervising routine servicing and major mechanical repairs, including interventions performed directly on the top drive while installed in the mast, when conditions and approved procedures permit. Daily washpipe lubrication checks, supported by a digital register tracking accumulated rotation hours from drilling data.

Drilling Data Acquisition & Real-Time Monitoring Systems

Practical experience using Electronic Drilling Recorders (EDR) and real-time drilling data acquisition systems, including PASON and NOV technologies, to monitor and interpret drilling parameters such as weight on bit (WOB), hook load, rate of penetration (ROP), rotary speed (RPM), torque, standpipe pressure, flow rate, pump strokes, mud pit levels and drilling depth. Analyse real-time and historical trends to identify abnormal operating conditions, support troubleshooting and verify equipment performance. Use recorded rotation hours and operating data to maintain digital equipment-service records, including top drive washpipe lubrication and maintenance tracking. Pay close attention to sensor accuracy and data reliability when assessing drilling conditions.

Drill String, BHA and Downhole Equipment Control

Technical review of BHA configuration and the identification, serial numbers and traceability of all downhole components, including DP, HWDP, drill collars, stabilizers, near-bit equipment, bits, crossovers and subs. Scrupulous inspection and tracking of tubulars and assemblies before running in hole. Practical understanding of drilling sequences, tripping and downhole operational constraints.

Fishing Equipment and Crew Preparedness

Working knowledge of overshots, grapple/diameter selection, extension guides, magnets, impression blocks, junk baskets, mechanical and hydraulic jars and bumpers. Maintain relevant equipment datasheets at the rig and conduct operational briefings with drillers, assistant drillers and supervisors to review tool function, operating limits and contingency use.

Well Control Readiness and BOP Systems

Plan theoretical and written well-control drills, including kill-sheet preparation based on current well depth and measurement of crew response times. Detailed knowledge of BOP stack components, function testing, fault diagnosis and field maintenance within safe, authorized operating conditions. Personally involved in field repair of a threaded-cover Hydril annular BOP, including packing-element replacement,

and repair of a double Shaffer ram BOP, including seal kits, front seals and top seals. Experience with hydraulic valves, connectors and control lines. Familiar with accumulator pre-charge checks using nitrogen, bladder replacement, four-way valves, pressure regulators and hydraulic-line repairs.

Drillmec Mud Pumps and Consumables

Component-level knowledge of Drillmec triplex mud pumps and both preventive and corrective maintenance. Monitor pump condition and optimize consumable life through drilling-operation checks and electronic tracking of liners, pistons, valves, seats, springs, guides and seals. Use recorded consumption and operating conditions to inform maintenance planning, inventory and cost control.

Mud Circulation, Solids Control and Gas Handling

Practical knowledge of mud tanks, fluid circulation circuits, vacuum degassers and atmospheric mud-gas separators, including operating checks and evaluation of suitability for expected flow and gas-handling demands. Experience with MI SWACO shale shakers and screen selection, balancing solids separation against fluid losses, screen wear and operating costs. Familiar with integrated desander/desilter/shaker packages (3-in-1), hydrocyclone selection, cut-point considerations and changing operating conditions.

Directional Drilling, Cementing and Recovery Operations

Operational familiarity with downhole motors, rotary steerable systems, Schlumberger PowerDrive and TeleScope systems, and MWD/LWD data interfaces. Understanding of tool response and downhole operating behavior, back-off operations, sidetracking and cementing arrangements, including stinger and plug/valve configurations. Coordination of specialized service providers and technical briefings; specialist directional tool operation is not claimed.

Automation, Power and Hydraulic Systems

Strong working knowledge of PLC-controlled rig functions, hydraulic and electrical schematics, hydraulic power units and rig equipment interfaces. Operational and maintenance familiarity with Caterpillar diesel power systems used on Drillmec HH rigs. Apply system-level fault analysis to identify likely causes, prioritize maintenance and reduce avoidable downtime.

Technical Leadership and Knowledge Transfer

Train and brief rig crews on component function, correct operation, equipment limitations and the consequences of misuse. Promote early detection of abnormal conditions, accurate maintenance records, structured troubleshooting and coordination between operations, maintenance and third-party specialists. Decisions on intrusive repairs, pressure-containing equipment and return to service remain subject to approved procedures, competent authorization and required testing.

Technical note: Equipment ratings, specific repair methods and maintenance intervals are rig/model dependent and should be checked against OEM manuals, equipment records and approved site procedures before operational use. This portfolio summarizes personal experience; it is not an operating instruction.

HSE Leadership, Safety Culture & Operational Discipline

I place HSE at the centre of rig management and maintain close, continuous cooperation with the HSE department. For me, reliable drilling performance depends on two inseparable priorities: preserving the health and safety of everyone working on the rig, and maintaining the integrity, reliability and efficiency of the equipment. Safe execution is an essential part of service quality, not a separate administrative exercise.

Permit-to-Work and Task-Specific Controls

Work with HSE personnel and supervisors to ensure that each activity is assessed, authorised and performed in accordance with applicable procedures and permit-to-work requirements. Particular attention is given to working at height, confined-space entry, welding and other hot work in hazardous or non-hazardous areas, isolation requirements, and correct selection and use of personal protective equipment (PPE).

Training, Competency and Crew Engagement

Promote initial and refresher training so that personnel understand equipment, task-specific hazards, operating procedures and their individual responsibilities. Encourage active participation by drillers, assistant drillers, supervisors, maintenance personnel and supporting crews, with practical explanations linked to real rig activities.

Pre-Job Briefings and Operational Coordination

Require pre-job meetings and toolbox talks before activities begin, particularly for non-routine or higher-risk work. Review the sequence of operations, personnel responsibilities, equipment readiness, interfaces between teams, relevant permits and the controls needed to prevent incidents and operational disruption.

Audits, Behavioural Observations and Continuous Improvement

Cooperate with HSE on scheduled inspections, formal audits and behavioural safety observations. Use findings to improve work practices, reinforce procedural compliance and identify training needs. Treat behavioural audits as opportunities for constructive improvement, not merely as a checklist.

Fitness for Duty and Consistent Application of Rules

Support unannounced, random alcohol testing in accordance with company policies and applicable local requirements, with the same rules applying to everyone, including myself. Promote consistent enforcement of HSE expectations regardless of role or seniority.

Equipment Integrity and Safe Maintenance

Coordinate operational and maintenance teams to ensure equipment remains serviceable and interventions are planned and executed under appropriate controls. Apply the same safety discipline to routine maintenance, major mechanical repairs, simultaneous operations and urgent field interventions, without allowing production pressure to override safe working conditions.

Management principle: Good HSE performance protects people, equipment and operational continuity. It is sustained through competent crews, clear procedures, active supervision, consistent controls and practical learning from field observations.